SQL TASK QUESTIONS

BASIC LEVEL

1. Create a new database named company_db.
2. Use the created database.
3. Create a table departments and add 10 values for that

```
CREATE TABLE departments (
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(50),
    location VARCHAR(50)
);
```

4. Create a table employees and add 10 values for that

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    gender CHAR(1),
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);
```

5. Create a table projects and add 10 values for that

```
CREATE TABLE projects (
    project_id INT PRIMARY KEY AUTO_INCREMENT,
    project_name VARCHAR(100),
    start_date DATE,
    end_date DATE,
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id));
```

6. Create a table employee_projects and add 10 values for that

```
CREATE TABLE employee_projects (
    emp_id INT,
    project_id INT,
    assigned_date DATE,
    role VARCHAR(50),
    PRIMARY KEY (emp_id, project_id),
    FOREIGN KEY (emp_id) REFERENCES employees(emp_id),
    FOREIGN KEY (project_id) REFERENCES projects(project_id)
);
```

SELECT

7. Select all employee names.
8. Select employees earning more than 60000.
9. Select first 5 employees.

WHERE CLAUSE

10. Display employees details in department 3.
11. Display Female employees details earning above 55000
12. Display employees hired after 2021.

ORDER BY CLAUSE

13. Sort employees by salary descending.
14. Sort departments alphabetically.
15. Sort employees by hire_date ascending.

AGGREGATE FUNCTIONS

16. Find total number of employees.
17. Find average salary.
18. Find maximum salary.

GROUP BY CLAUSE

19. Find total employees in each department.
20. Find average salary by gender.
21. Find min and max salary per department.

HAVING CLAUSE

22. Show departments with more than 2 employees.
23. Show Departments with average salary above 60000.
24. Show Gender with average salary greater than 58000.

JOINS

25. Get employee name with department name.
26. Show all projects with department names.
27. Show employees and their assigned project names.

UPDATE

28. Increase salary by 10% for IT department employees.
29. Change any one employee department.
30. Update location of Sales department to Noida

SUBQUERIES

31. Find employees earning above average salary.
32. Find employees working in 'IT' department.
33. Find department names where employees earn more than 70000.

MEDIUM LEVEL

1. Display full names and department names of all employees working in the 'IT' department.
2. Find the top 3 highest-paid employees and show their department names.
3. Show the total salary expense of each department.
4. Find departments where more than 2 employees work.
5. List all employees who joined before the year 2021.
6. Find the department that pays the highest average salary.
7. Display all project names handled by the 'Finance' department.
8. List employee names, project names, and their roles.
9. Find all employees who are not assigned to any project.
10. Show all departments and the number of projects handled by each.
11. Find the employee(s) who earn the maximum salary.
12. Find employees who work in departments located in 'Delhi'.
13. List the number of male and female employees in each department.
14. Find employees whose salary is above the average salary of their department.
15. Show all employees with their department, project name, and project's start date — even if they are not assigned to any project.

SAMPLE TABLE VALUES

- INSERT INTO departments (dept_name, location) VALUES

  ('HR', 'Delhi'),

  ('Finance', 'Mumbai'),

  ('IT', 'Bangalore'),

  ('Sales', 'Chennai'),

  ('Marketing', 'Pune'),

  ('Operations', 'Hyderabad'),

  ('Support', 'Kolkata'),

  ('Admin', 'Delhi'),

  ('Logistics', 'Jaipur'),

  ('Legal', 'Ahmedabad');

- INSERT INTO employees (first_name, last_name, gender, salary, hire_date, dept_id) VALUES

  ('Amit', 'Sharma', 'M', 60000, '2020-02-15', 3),

  ('Priya', 'Kumar', 'F', 55000, '2019-11-20', 1),

  ('Rohit', 'Verma', 'M', 72000, '2018-07-12', 2),

  ('Sneha', 'Rao', 'F', 65000, '2021-04-05', 3),

  ('Vikas', 'Patil', 'M', 48000, '2022-01-17', 4),

  ('Neha', 'Singh', 'F', 58000, '2021-10-22', 5),

  ('Arjun', 'Mehta', 'M', 75000, '2019-06-30', 2),

  ('Kiran', 'Das', 'F', 62000, '2020-09-25', 1),

  ('Deepak', 'Joshi', 'M', 53000, '2022-07-11', 6),

  ('Pooja', 'Nair', 'F', 50000, '2023-02-01', 7);

- INSERT INTO projects (project_name, start_date, end_date, dept_id) VALUES

  ('Payroll System', '2020-01-01', '2020-12-31', 2),

  ('Employee Portal', '2021-03-01', '2021-10-15', 3),

('CRM Development', '2019-05-20', '2020-02-28', 4),

('Recruitment Drive', '2022-06-01', '2022-12-01', 1),

('Inventory Tracker', '2021-07-15', '2022-05-30', 6),

('Ad Campaign', '2023-01-10', '2023-08-01', 5),

('Client Support Tool', '2020-04-12', '2021-02-18', 7),

('Legal Docs System', '2021-09-01', '2022-03-15', 10),

('Transport Scheduler', '2022-02-25', '2023-01-01', 9),

('Admin Automation', '2021-11-11', '2022-07-20', 8);

- INSERT INTO employee_projects VALUES

  (1, 2, '2021-03-05', 'Developer'),

  (2, 4, '2022-06-03', 'HR Coordinator'),

  (3, 1, '2020-01-05', 'Finance Analyst'),

  (4, 2, '2021-03-10', 'Tester'),

  (5, 3, '2019-05-25', 'Sales Executive'),

  (6, 6, '2023-01-12', 'Marketing Manager'),

  (7, 1, '2020-01-10', 'Finance Lead'),

  (8, 4, '2022-06-05', 'Recruiter'),

  (9, 5, '2021-07-20', 'Operations Executive'),

  (10, 7, '2020-04-15', 'Support Engineer');